FYS5419 Project 2

Report Template

Your Name
University of Oslo

Spring 2026

Abstract

Write a concise summary of the problem, method, main numerical results, and conclusion. Replace this template text after you have chosen one of the Project 2 alternatives.

Contents

1 Introduction

Explain the scientific problem and why the selected quantum algorithm or quantum machine learning model is relevant. State clearly which Project 2 alternative you chose.

Chosen path. TODO: Select exactly one of the following paths and remove the unused material from the final report:

1. Quantum Fourier Transform and Quantum Phase Estimation.
2. Quantum Machine Learning with the Iris data set.
3. Variational Quantum Boltzmann Machines.
4. Adaptive VQE / eigenvalue problems.

2 Theory

2.1 Quantum Fourier Transform path

For the QFT path, start from the definition

$$|j\rangle \mapsto \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i jk/N} |k\rangle, \tag{1}$$

and show that the transformation is unitary. Include the one-, two-, and three-qubit matrices, then derive the gate decomposition and inverse QFT.

2.2 Quantum Phase Estimation path

Describe the eigenvalue equation

$$U|u\rangle = e^{2\pi i\phi}|u\rangle \tag{2}$$

and how the inverse QFT extracts the phase estimate $\tilde{\phi}$. Discuss the relation between phase and energy for a time-evolution operator $U = \exp(-iHt)$.

2.3 Quantum Machine Learning path

For the QML path, describe the feature map, ansatz, measurement rule, loss function, and parameter-shift gradient. For binary classification, define the prediction $f(\boldsymbol{x}; \boldsymbol{\theta})$ and cross-entropy loss.

3 Numerical methods and implementation

Describe the code structure, algorithms, and validation tests. Put reusable code in `src/fys5419_project_2/`; scripts used to generate results can go in `scripts/`. Include pseudocode where useful.

3.1 Reproducibility

Record the Python version, package versions, random seeds, and all parameters used to generate the final figures and tables.

4 Results

Place final figures in `results/figures/` and final tables in `results/tables/`. Every table and figure should have a label, caption, and clear axes/units where applicable.

Table 1: Example results table. Replace with your final numerical results.

Method	Error	Runtime
Reference	TODO	TODO
Project implementation	TODO	TODO

5 Discussion

Interpret the results. Discuss accuracy, numerical stability, scaling, and limitations. For comparisons, explain what was held fixed and what was varied.

6 Conclusion

Summarize what was implemented, what was validated, and what you learned. Mention possible improvements or extensions.

Acknowledgements and collaboration

State collaborators, division of work, and any AI/tool assistance according to the course requirements.

A Code usage

Give the exact commands used to reproduce the final results. Example:

```
python scripts/run_qft_reference.py
make pdf
```

B Additional derivations

Use the appendix for derivations or extra results that would interrupt the main text.

References

[1] Robert Hundt, *Quantum Computing for Programmers*, Cambridge University Press, 2022.

[2] Michael A. Nielsen and Isaac L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2000.

[3] CompPhysics, *QuantumComputingMachineLearning course repository*, https://github.com/CompPhysics/QuantumComputingMachineLearning.